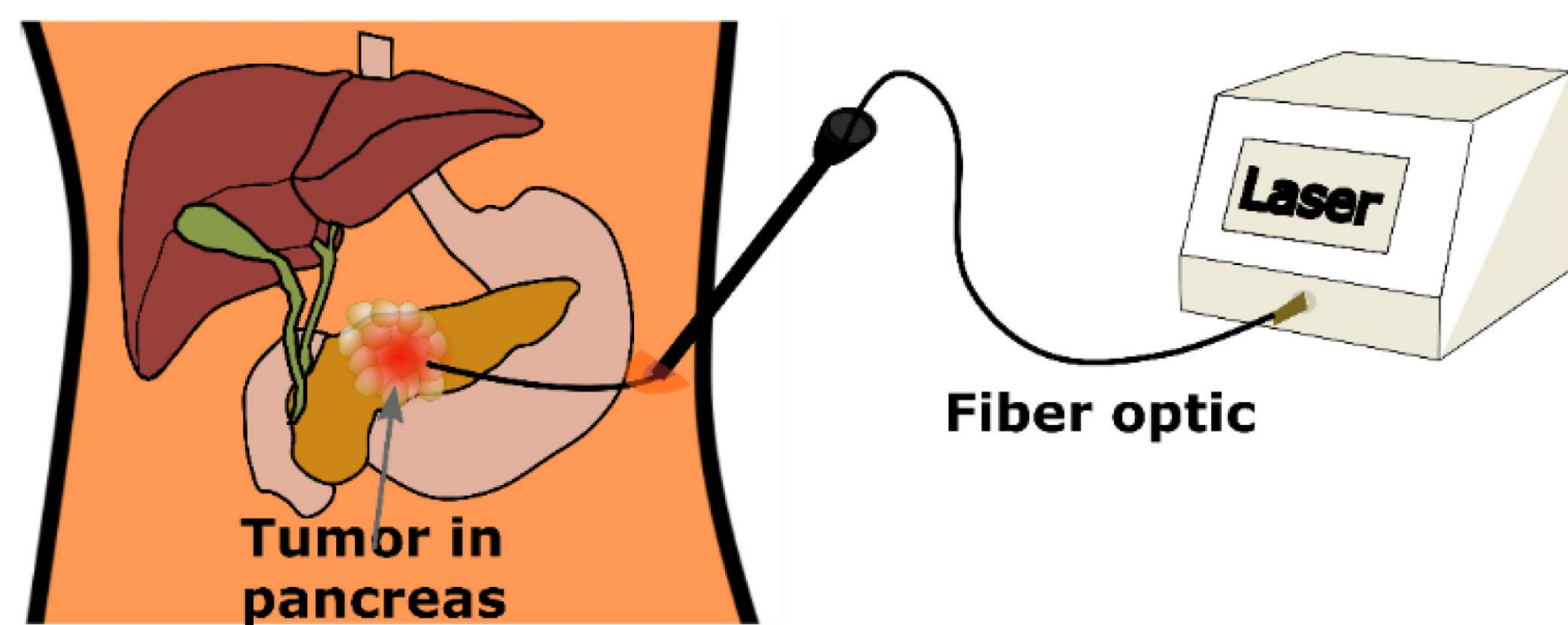


Background:

- Radiomics: science of extracting structured features from CT scans
- It is hard to measure only by the human eye the subtle textural changes in CT that reflect treatment response.
- Radiomics used to assess PDT damage to pancreatic tumors
- PDT is alternative treatment of chemotherapy for patients with pancreatic cancer by shining laser light at tumor cells
- We are using mRMR and SVM to rank top features and classify patients into responders and non-responders

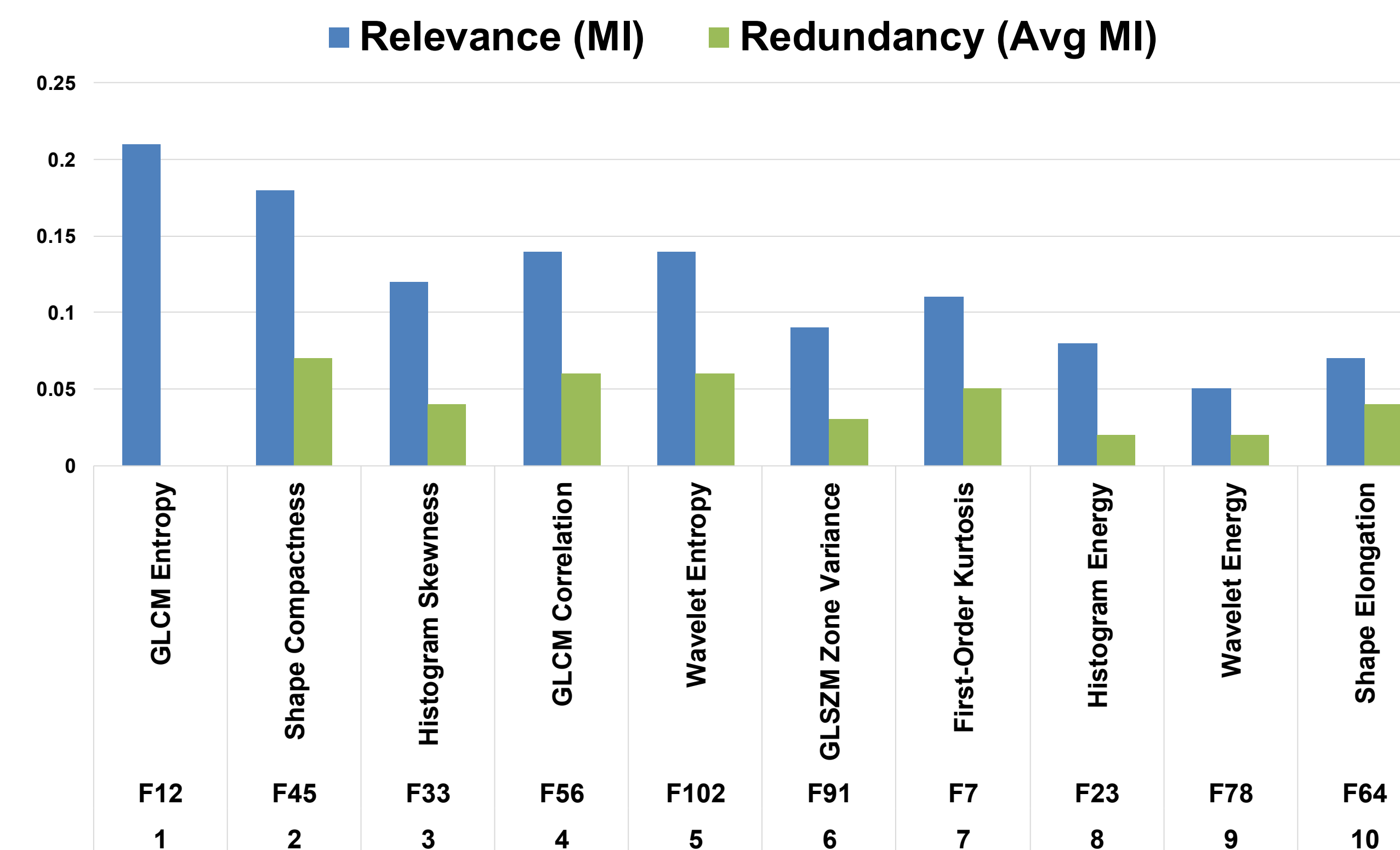


(Karimnia V et al. 2021)

$$f(x) = \text{sign}(w \cdot x + b)$$

- $w \cdot x + b$ computes the signed distance of a data point from the decision boundary (hyperplane)
- $\text{sign}()$ converts this distance into a binary classification: +1 (responder) or 0 (non-responder)
- The hyperplane $w \cdot x + b = 0$ is the optimal boundary that maximally separates the two classes in feature space

Features through mRMR selection ranked

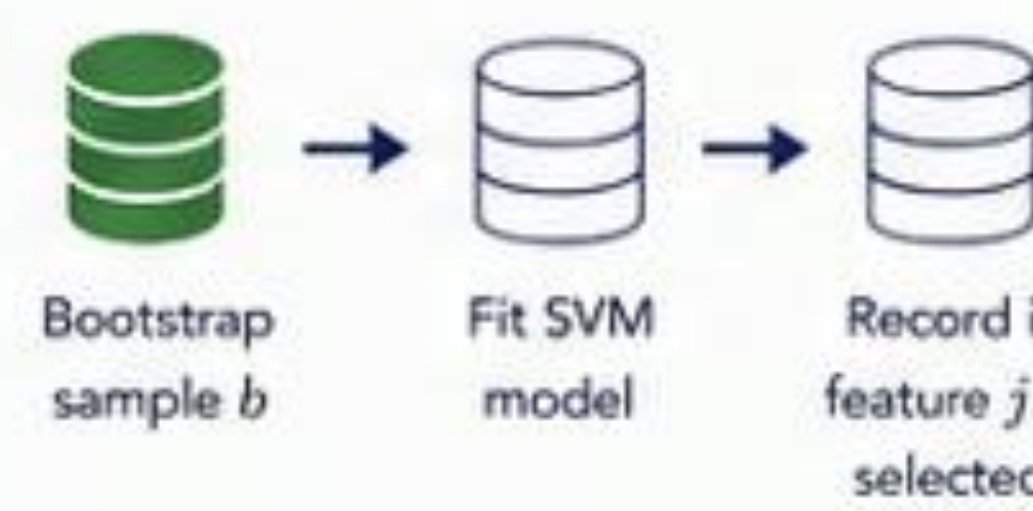


Evaluate stability of each feature across resampling (e.g., bootstrapping).

For each feature j ($j = 1 \dots 20$):

Resampling Procedure

Repeat B times (e.g., $B = 100$):



Stability Score

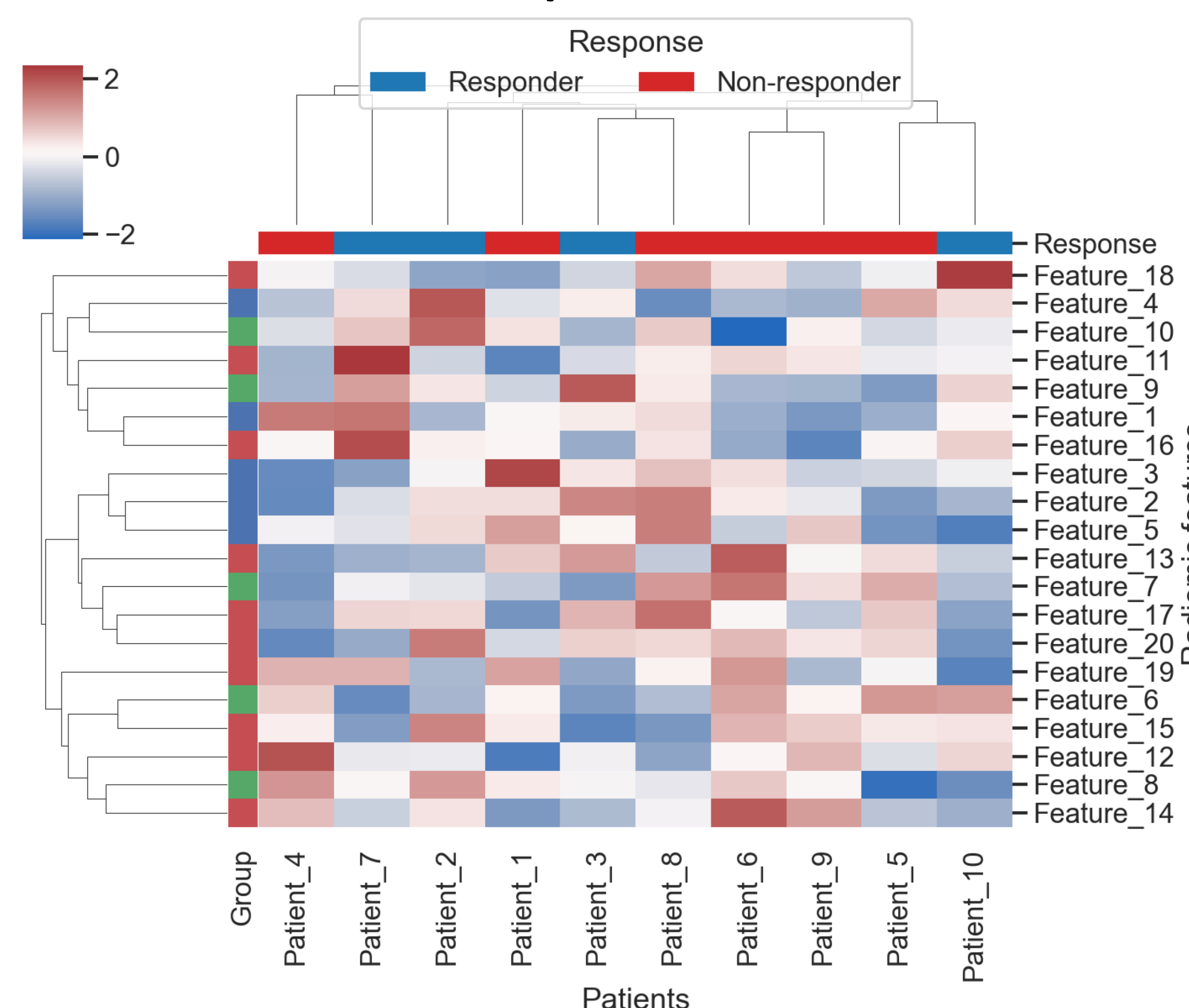
$$S_j = \frac{\# \text{ of times feature } j \text{ is selected}}{B}$$

$$0 \leq S_j \leq 1$$

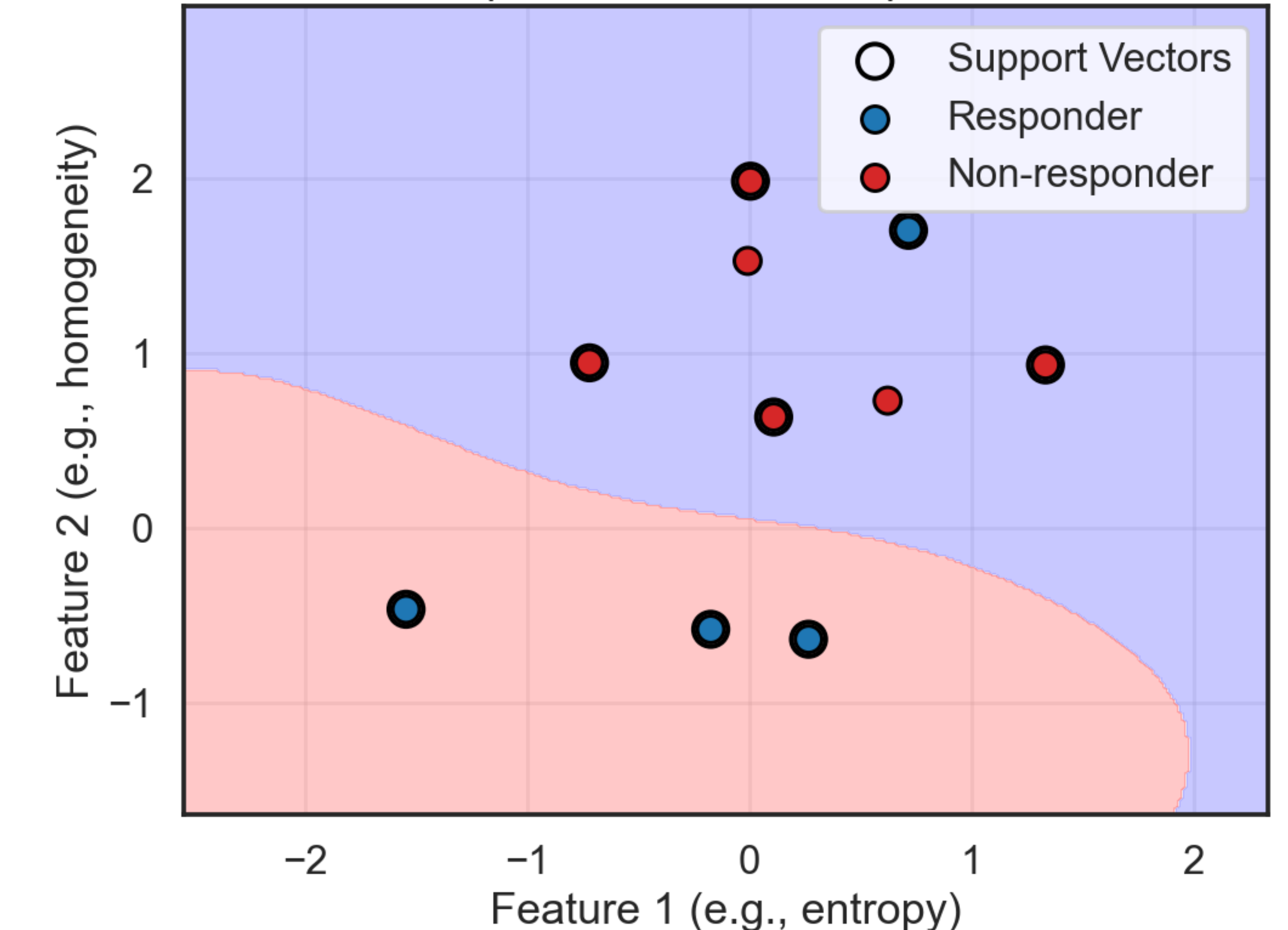
Low Stability

High Stability

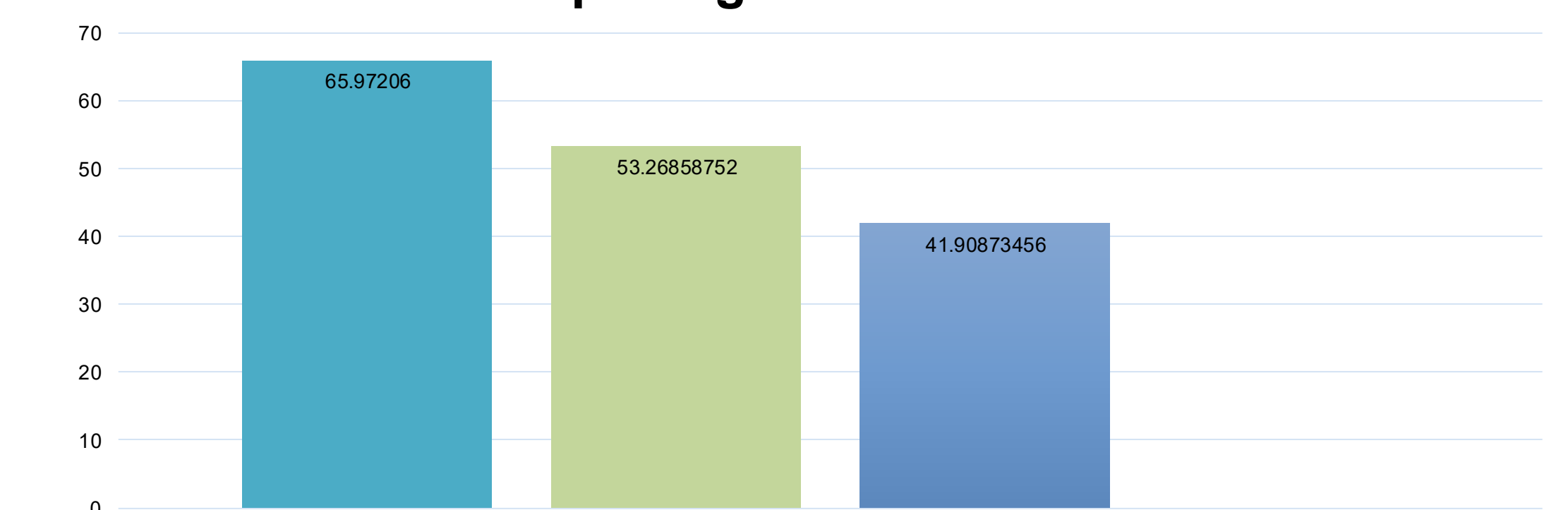
Predicted Response based on SVM:



2D Scatter Plot with SVM Decision Boundary Responders vs Non-Responders



Morphological Volume cm3



Results/Analysis:

- Data fits the limited ground truth correctly
- Radiologist agree that top features Entropy and Compactness are reasonable.
- Overall, Morphological Volume shown to decrease

Limitations:

- Semi supervised training
- Lack of hyperparameterization
- Larger dataset to further validate SVM performance
- Current analysis uses the whole pancreas, not just the tumor
- **Next step:** segment tumor-only regions with radiologist input (supervised) and large dataset

References:

- Karimnia V, Slack FJ, Celli JP. Photodynamic Therapy for Pancreatic Ductal Adenocarcinoma. *Cancers*. 2021; 13(17):4354. <https://doi.org/10.3390/cancers13174354>
- Vincent P, Maeder ME, Hunt B, et al. CT radiomic features of photodynamic priming in clinical pancreatic adenocarcinoma treatment. *Phys Med Biol*. 2021;66(17):10.1088/1361-6560/ac1458. Published 2021 Aug 23. doi:10.1088/1361-6560/ac1458

